Hints: Linear algebra – 4. Functions of matrices and matrix equations

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Kronecker route: $X \mapsto AX - XB$ is a linear operator on M_n. Under vec its matrix is $E \otimes A - B^T \otimes E$. Triangularise A and B (hence B^T) simultaneously in the two tensor factors to see that its eigenvalues are $\lambda_i - \mu_j$. Unique solvability for every C means this operator is invertible.

Hint 2. Integral route: let the contour Γ surround $\operatorname{Spec} A$ but not $\operatorname{Spec} B$. Use $A(zE - A)^{-1} = z(zE - A)^{-1} - E$ and the analogous identity for B to compute $AX - XB$ for the given integral. Two integrals remain: $\frac{1}{2\pi i} \oint (zE - A)^{-1} dz = E$ and $\oint (zE - B)^{-1} dz = O$.

Hint 3. Conversely, if λ is a common eigenvalue, $X = uw^T$ with $Au = \lambda u$, $w^T B = \lambda w^T$ is a non-zero solution of $AX - XB = O$. For the last claim, take $X = E_{ij}$ (matrix units) in $AX = XA$.

Problem 2

Hint 1. The two matrices must not commute. Try nilpotent ones, whose exponentials are trivial to compute: $A = E_{12}$, $B = E_{21}$.

Hint 2. $e^A e^B = (E + E_{12})(E + E_{21})$ has integer entries. $A + B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ squares to E, so $e^{A+B} = \cosh 1 \, E + \sinh 1 \, (A + B)$.

Problem 3

Hint 1. The target is not diagonalisable. If A had two distinct eigenvalues, A and hence $\sin A$ would be diagonalisable. If A is scalar, so is $\sin A$.

Hint 2. So $A = \lambda E + N$ with $N \neq O$, $N^2 = O$, and $\sin A = \sin \lambda \, E + \cos \lambda \, N$. The eigenvalue condition gives $\sin \lambda = 1$, hence $\cos \lambda = 0$ and $\sin A = E$. Answer: no such A.

Problem 4

Hint 1. A matrix is nilpotent if and only if all its eigenvalues are 0. By the spectral mapping principle, the eigenvalues of $p(e^{AB})$ are $p(e^\lambda)$ over the eigenvalues λ of AB.

Hint 2. AB and BA have the same characteristic polynomial, so the same eigenvalues. The statement is true.

Problem 5

Hint 1. The left side is $L(X) = (I + \mathcal{N})(X)$, where $\mathcal{N}(X) = AX + XA^2$ is the sum of two commuting nilpotent operators on $M_n(\mathbb{R})$ (left multiplication by A and right multiplication by A^2). So $\mathcal{N}$ is nilpotent and L is invertible.

Hint 2. $X = L^{-1}(A) = \sum_k (-1)^k \mathcal{N}^k(A)$. Compute $\mathcal{N}(A) = A^2$ and $\mathcal{N}^2(A) = 0$. Answer: $X = A - A^2$.

Problem 6

Hint 1. Write the left side as a linear operator on matrices: $L = \alpha_{00} I + \sum_{(i,j)\neq(0,0)} \alpha_{ij}\, \mathcal{L}_A^i \mathcal{R}_A^j$, where $\mathcal{L}_A X = AX$ and $\mathcal{R}_A X = XA$.

Hint 2. $\mathcal{L}_A$ and $\mathcal{R}_A$ commute and are nilpotent, so the second sum is nilpotent. A non-zero scalar plus a nilpotent operator is invertible.

Problem 7

Hint 1. Any solution commutes with $A = X^2$. First find one obvious square root that is a polynomial in A: with $M = A - \frac{1}{4}E$ (so $M^2 = O$), try $F = \frac{1}{2}E + M$.

Hint 2. $F = A + \frac{1}{4}E$ satisfies $F^2 = A$ and is invertible. For any solution X, put $S = XF^{-1}$. Since X and F commute, $S^2 = X^2 F^{-2} = E$, and S commutes with A.

Hint 3. Conversely, every involution S commuting with A gives a solution $X = S(A + \frac{1}{4}E)$. Describe these S: either $S = \pm E$, or S splits $\mathbb{C}^4$ into two A-invariant 2-dimensional subspaces on which A is a single Jordan block. In the basis of the sheet, a matrix commuting with A has 2×2 blocks $aE + bN$, $N = E_{12}$.